

# Mohamad Chehade

PH.D. STUDENT · ELECTRICAL AND COMPUTER ENGINEERING

*The University of Texas at Austin*

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## Education

### The University of Texas at Austin

*Austin, Texas*

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING

*Aug. 2023 – Present*

- Advisor: Dr. Hao Zhu
- Research Interests: Large Language Model Alignment, Time Series Foundation Models, Risk-aware Transfer Learning, Learning for Optimization and Control, Neural Network Verification and Certified Robustness

### The University of Texas at Austin

*Austin, Texas*

MASTER OF SCIENCE IN ENGINEERING

*Aug. 2023 – May 2026*

- Advisor: Dr. Hao Zhu
- Relevant Courses: Reinforcement Learning, Generative AI, Statistical Machine Learning, Applied Machine Learning, Probability and Stochastic Processes, Applied Stochastic Processes, Learning-based Optimal Control, Convex Optimization, Energy Optimization and Operation

### American University of Beirut

*Beirut, Lebanon*

B.ENG. IN ELECTRICAL AND COMPUTER ENGINEERING

*Aug. 2019 – Jun. 2023*

- GPA: 4.0/4.0 | Minor in Mathematics
- Final Year Project: Optimal Power Flow via Machine Learning (Advisor: Dr. Rabih Jabr)
- Undergraduate Research: Microgrid Sizing using Ordinal Optimization (Advisor: Dr. Sami Karaki)

## Experience

### Argonne National Laboratory - Grid Optimization and Infrastructure Group

*Remote*

RESEARCH AIDE – MENTORS: DR. FENG QIU & DR. WEI GAO

*May 2026 – Aug. 2026*

- Developing decision-aware time series foundation models that align forecasting with downstream decision-making and optimization.

### Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group

*Remote*

GRADUATE RESEARCH ASSISTANT (GRA) – MENTOR: DR. RUSSELL BENT

*May 2025 - Aug. 2025*

- Extended neural network verification ideas toward robustness training and certified robustness
- Developed a training algorithm that leverages verifiable input regions to improve certified robustness of neural networks

### Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group

*Los Alamos, NM*

GRADUATE RESEARCH ASSISTANT (GRA) – MENTORS: DR. WENTING LI, DR. BRIAN BELL

*Jun. 2024 - Aug. 2024*

- Worked on the verification of neural networks in physical and safety-critical systems
- Developed two algorithms for determining large verifiable input regions of neural networks

### University of Connecticut - Center for Clean Energy Engineering - PEARL Lab

*Storrs, CT*

RESEARCH INTERNSHIP – ADVISOR: DR. ALI BAZZI

*Jun. 2022 - Aug. 2022*

- Collaborated with Ward Leonard, a leading industrial motor manufacturer, on fault-tolerant power electronics
- Developed a fault diagnosis algorithm for power electronic inverters using combinational logic
- Optimized and constructed the inverter circuit for high-power testing using mixed-integer linear programming

## Publications

### PUBLISHED / ACCEPTED

**Chehade, M., & Zhu, H. (2026).** NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS 2026)*. arXiv:2509.21668 | Code

**Chehade, M., & Zhu, H. (2026).** Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. Accepted to 2026 IEEE Power & Energy Society General Meeting (PESGM 2026).

Cho, Y.-H., **Chehade, M.**, Al-Janahi, F., Lim, S., Mohammadi, J., & Zhu, H. (2026). Carbon-aware Market Participation for Building Energy Management Systems. Accepted to 2026 IEEE Power & Energy Society General Meeting (PESGM 2026).

- Chehade, M.**, Li, W., Bell, B. W., Bent, R., Kazi, S. R., & Zhu, H. (2025). LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2408.08824 | OpenReview
- Chehade, M.**, Ghosal, S. S., Chakraborty, S., Reddy, A., Manocha, D., Bedi, A. S., & Zhu, H. (2025). Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2505.23729 | OpenReview
- Chehade, M.**, & Karaki, S. (2025). BOOST: Microgrid Sizing Using Ordinal Optimization. *2025 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, pp. 1–4. doi:10.1109/TPEC63981.2025.10907217.
- Chehade, M.**, Cho, Y.-H., Chinchali, S., & Zhu, H. (2024). Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *2024 56th North American Power Symposium (NAPS)*, El Paso, TX, USA, pp. 1–5. doi:10.1109/NAPS61145.2024.10741791.

## UNDER REVIEW

- Chehade, M.**, Bedi, A. S., Chakraborty, S., Zhang, A., & Zhu, H. (2026). Test-Time Risk Adaptation with Mixture of Agents. Under review at *NeurIPS 2026*. arXiv:2408.08812

## Skills

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- **Programming Languages:** Python, MATLAB, C/C++, SQL, R
- **ML / Scientific Computing:** PyTorch, NumPy, SciPy, Scikit-learn, Matplotlib, Gurobi, MATPOWER
- **Engineering & Power Systems Tools:** Simulink, SPICE, HOMER, PVSyst, LabVIEW
- **Tools & Workflow:** Git,  $\LaTeX$ , Linux, Overleaf, Microsoft Office Suite
- **Languages:** English (fluent), French (fluent), Arabic (native)

## Reviewer

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|---------------------|----------------------------------------------------------------------|
| Sep. 2023 - Present | <b>IEEE Transactions on Smart Grid</b>                               |
| July 2024 - Present | <b>Asilomar Conference on Signals, Systems, and Computers</b>        |
| Dec. 2024 - Present | <b>Texas Power and Energy Conference (TPEC)</b>                      |
| June 2025 - Present | <b>Hawaii International Conference on System Sciences (HICSS)</b>    |
| Sep. 2025 - Present | <b>IEEE Systems Journal</b>                                          |
| Oct. 2025 - Present | <b>International Conference on Learning Representations (ICLR)</b>   |
| Feb. 2026 - Present | <b>International Conference on Machine Learning (ICML)</b>           |
| May 2026 - Present  | <b>IEEE Power Engineering Letters</b>                                |
| June 2026 - Present | <b>Conference on Neural Information Processing Systems (NeurIPS)</b> |

## Awards & Honors

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|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026           | <b>Third Place, ASME CIE Student Hackathon ASME IDETC/CIE 2026</b>                                                                                 |
| 2024           | <b>Best Graduate Presentation Award at NAPS 2024</b>                                                                                               |
| 2023 - Present | <b>Cockrell School of Engineering Fellowship</b>                                                                                                   |
| 2023           | <b>Mohamad Ali Safieddine Award for Academic Excellence</b> for ranking first across the AUB Maroun Semaan Faculty of Engineering and Architecture |
| 2023           | <b>ECE Distinguished Graduate Award</b> for ranking first among ECE graduates                                                                      |
| 2023           | <b>Exceptional ECE Final Year Project Award</b> Power and Energy Systems                                                                           |
| 2021           | <b>Best Pitch Award EPFL Tech4Impact Summer School</b>                                                                                             |
| 2019 - 2023    | <b>Dean's Honor List</b> AUB Maroun Faculty of Engineering and Architecture - every given semester                                                 |